

high voltage tank item table

No.	Type	Item	Single-material Part / Assembly	Function	Composition	Materials	Need Rationale Source / Page / Verbatim Quote	Material Source / Page / Verbatim Quote
HV-1	Structure	High-voltage tank enclosure	Assembly	Hold the oil-filled HV transformer, rectifier stack, output filter capacitors, insulating fluid, and tank-side interfaces.	Tank body, lid, cover/flange, grounding terminal, oil- compatible seals, fill/drain interfaces, internal mounting points.	304/316L stainless steel or aluminum alloy; oil- compatible seals; copper or stainless grounding terminals.	This is a BINP conference paper describing a 60 kV / 15 kW high-voltage source for electron- beam equipment. Page p.2. Quote: "oil- filled tank."	This is an Eaton product catalog for liquid-filled pad-mounted transformers; it is used as a reference for tank/cabinet hardware. Page p.4. Quote: "Stainless steel 1- hole ground pads."
HV-2	High-voltage Transformer	Transformer insulating fluid	Single material	Provide dielectric insulation in the HV tank, suppress partial discharge, and help heat transfer.	Insulating liquid only.	Silicon oil / silicone transformer oil / PDMS silicone fluid.	This is the same BINP conference paper on the 60 kV / 15 kW high-voltage source; it directly specifies the insulating fluid. Page p.2. Quote: "silicon oil."	This is the same BINP high-voltage source paper; it directly specifies the material used as the tank fluid. Page p.2. Quote: "silicon oil."
HV-3	High-voltage Transformer	Sectioned HV step-up transformer	Assembly	Step up the high-frequency AC input from power supplies to electron-gun accelerating voltage.	Magnetic core, primary winding, sectioned secondary winding, interlayer insulation, bobbin/support, terminals, oil- immersed mounting.	Copper winding or copper foil; magnetic core material to be selected; polyimide, Nomex, pressboard, oil- paper insulation.	This is the BINP conference paper describing the high- voltage volume and its sectioned transformer architecture. Page p.2. Quote: "winding, half-bridge rectifier and output filter capacitors."	This is Weidmann's transformer insulation component page describing insulation parts used inside transformer winding assemblies. Web page. Quote: "transformer winding assemblies."
HV-4	High-voltage Transformer	HV rectifier stack	Assembly	Rectify stepped-up AC into DC accelerating voltage.	Series/parallel HV diodes, equalizing resistors, terminals, insulating supports, oil- side mounting.	HV silicon diodes; epoxy or ceramic encapsulation; metal terminals; HV equalizing resistors.	This is the BINP high- voltage source paper describing the contents of each transformer section. Page p.2. Quote: "half-bridge rectifier."	This is TT Electronics' high- voltage resistor application note; it supports use of high-voltage resistors for balancing in HV circuits, while diode stack packaging still needs supplier confirmation. Page p.1. Quote: "Voltage balancing resistors."
HV-5	High-voltage Transformer	HV output filter capacitor	Assembly	Reduce output ripple and stabilize electron-beam energy.	HV capacitors, series/parallel connections, equalizing or discharge resistors, terminals, oil- side insulating supports.	HV ceramic capacitors or polypropylene film capacitors; metal terminals; ceramic/polymer supports.	This is the BINP high- voltage source paper; it states that each section includes output filter capacitors. Page p.2. Quote: "output filter capacitors."	This is the same BINP paper; it gives the electrical purpose of the filter capacitance. Page p.2. Quote: "40 kHz ripples less than 0,5%."
HV-6	Interface	Internal HV leads and terminals	Assembly	Electrically connect transformer secondary sections, rectifier stack, filter capacitors, and HV output bushing.	Internal conductors, oil- side terminals, connection straps, studs, rounded terminal caps.	Aluminum alloy conductor, OFHC copper, brass, stainless steel; optional nickel or silver plating.	This is a GE Vernova installation/operation manual for oil- immersed transformer high- current bushings; it explains how oil-side conductors connect to the transformer winding. Page p.3. Quote: "terminals for connection to the transformer winding."	This is the same GE Vernova bushing manual; it gives an example conductor/terminal material. Page p.3. Quote: "aluminium alloy casting."
HV-7	Structure	Transformer insulation spacers	Multiple parts; individual spacer can be single-material	Mechanically support HV transformer/rectifier/capacitor assemblies and maintain oil- side insulation distances.	Spacers, standoffs, clamps, support plates, insulating	Pressboard, oil- paper insulation, alumina ceramic, G10/G11, PEEK, PTFE, polyimide.	This is Weidmann's transformer insulation spacing- elements page; it describes spacing elements used inside	This is the same Weidmann insulation- components page; it supports insulation and

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					studs, locating pieces.		transformer windings. Web page. Quote: “radial separation between conductors.”	cooling functions. Web page. Quote: “support insulation integrity and cooling efficiency.”
HV-8	Interface	Tank-side HV output bushing / feedthrough / socket	Assembly	Bring HV from the oil-filled tank to the external HV cable while preserving insulation and sealing.	Central conductor, ceramic insulator, metal housing, brazed joint, oil-side terminal, cable socket, field grading/corona shield.	High-purity alumina ceramic; Kovar; 304/316 stainless steel; copper, nickel, molybdenum conductor options.	This is CeramTec’s ceramic-to-metal electrical feedthrough catalog; it gives voltage capability for high-voltage feedthroughs. Page p.36. Quote: “100 KV.” This is a GE Vernova oil-to-air bushing manual; it supports including field grading in the bushing assembly. Quote: “electrical gradient.”	This is CeramTec’s high-voltage feedthrough catalog; it lists typical feedthrough materials. Page p.33 / p.36. Quote: “Housing: Kovar,” “Insulation: Alumina ceramic,” and “Housing: 304 Stainless steel.”
HV-9	Interface	HV cable to fixed electron beam gun	Assembly	Transmit accelerating voltage from the HV tank to the fixed electron beam gun.	Central conductor, main dielectric, semiconductive layer, braided shield, outer jacket, cable termination.	Copper conductor; EPR or silicone dielectric; semiconductive layers; braided shield; polymer outer jacket.	This is a NASA EBF ³ patent describing the connection between the electron beam gun and the high-voltage power supply. Page p.8. Quote: “high Voltage power cable.”	This is Hivolt’s high-voltage cable and connector catalog covering X-ray / E-beam high-voltage cables. Page p.1. Quote: “EPR or Silicone dielectric” and “semiconductive layers, braided shield.”
HV-10	Protections and Interlocks	HV discharge / bleeder resistor chain	Assembly	Discharge HV capacitors after shutdown and reduce residual stored-energy hazard.	Series HV resistors, insulating supports, terminals, shielding/fixing hardware.	High-voltage thick-film resistor, metal-glaze resistor, ceramic body.	This is TT Electronics’ high-voltage resistor application note; it explains how bleeder resistors are used in high-voltage circuits. Page p.2. Quote: “discharge capacitors to safe voltage levels.”	This is the same TT Electronics application note; it describes high-voltage resistor technology and applications. Page p.1. Quote: “compact high voltage resistors is thick-film.”
HV-11	Control	HV voltage divider / output voltage sensing hardware	Assembly	Measure HV output and provide a scaled low-voltage feedback signal to controls.	HV resistor chain, low-voltage resistor, insulating column, shield, low-voltage signal connector.	Precision HV resistor chain; ceramic/polymer support; shielded cable.	This is the measured-channel table in the BINP high-voltage source paper; it lists output-voltage measurement. Page p.3. Quote: “Output voltage.”	This is TT Electronics’ high-voltage resistor application note; it explains the use of HV resistors in voltage dividers. Page p.4. Quote: “voltage dividers for the measurement.”
HV-12	Protections and Interlocks	HV output or return current monitor	Assembly	Measure output/return current for protection, beam-current monitoring, and fault detection.	Current shunt or current transformer, insulating mount, signal terminal, shielded lead.	Manganin shunt, metal foil shunt, current transformer core, Hall sensor, shielded cable.	This is the protection section of the BINP high-voltage source paper; it states that breakdown protection depends on output-current monitoring. Page p.3. Quote: “Breakdown protection controls output current.”	This is Isabellenhütte’s MANGANIN® material datasheet; it supports using Manganin for precision and shunt resistors. Page p.1. Quote: “standard material for precision, standard and shunt resistors.”
HV-13	Structure	Oil service, level, and pressure-relief hardware	Assembly	Support oil filling, draining, sampling, level check, thermal expansion, and pressure relief.	Fill port, drain port, sampling valve, level gauge, pressure relief valve, bleed/vent port, bellows or expansion bladder.	Stainless steel fittings, metal valves, FKM/fluorosilicone seals, metal bellows, level gauge or level sensor.	This is Eaton’s liquid-filled pad-mounted transformer catalog; it lists tank service hardware used on liquid-filled transformer tanks. Page p.4–5. Quote: “Drain valve with sampler,” “Automatic Pressure relief valve,”	This is the same Eaton catalog; it lists stainless tank/cabinet options and related service hardware. Page p.5. Quote: “Stainless steel tank base” and

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							and "Liquid level gauge."	"Liquid level gauge."
HV-14	Protections and Interlocks	Oil and HV transformer temperature sensing hardware	Assembly	Measure oil/transformer temperature and send protection signals to controls.	RTD/thermistor, oil probe, sealed feedthrough, input-current sensor, signal leads.	Platinum RTD, thermistor, 316 stainless steel sheath, ceramic insulation, shielded leads.	This is the protection/measured-channel section of the BINP high-voltage source paper. Page p.3. Quote: "Temperature of high-voltage transformer."	Material candidates are engineering selections based on common industrial temperature-sensor construction; this row still needs a specific supplier datasheet for final material lock.
HV-15	Protections and Interlocks	Tank grounding, shielding, and service interlock hardware	Assembly	Provide tank grounding, service safety, interlock hardware, and shielding.	Ground stud, grounding strap, enclosure shield, lid/service interlock switch, service panel terminal, manual discharge point.	Stainless steel enclosure, copper grounding strap, braided tinned copper, mechanical interlock switch.	This is Spellman's high-voltage power supply operation manual; it explains grounding and external interlock requirements for HV power hardware. Page p.14. Quote: "must be grounded." Page p.9. Quote: "External Interlock."	This is Hivolt's high-voltage cable catalog; it supports braided shielding as an HV hardware material family. Page p.1. Quote: "braided shield."